CLAIMPROMOTEBENCH: COMPILING POLICY DIFFERENCES INTO FAILURE-AWARE STATE-TRANSITION TESTS

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ABSTRACT

Model-mediated workflows can apply versioned policies to structured evidence before emitting state changes. Fixed-policy accuracy cannot show that a route used the active version, and scoring only valid outputs hides interface failure. We introduce CLAIMPROMOTEBENCH, a synthetic conformance-testing protocol that compiles evidence changes and executable policy differences into action-changing pairs. Migration adequacy separates changed-rule from reachable action-transition coverage; exact dynamic programs then select a minimum joint suite. In one authored migration, a rule-complete suite covers 6/10 reachable edges, whereas a graph-complete suite covers all ten, certifies two impossible directions, and yields a 10-pair gate covering 12 changed rules. Five frozen evaluations contain 420 pairs across two ontologies. Every scheduled service, syntax, schema, and semantic failure remains incorrect end to end. In hidden r6, Qwen returns 272/300 valid and 258/300 correct decisions, with 113/150 correct pairs. On an action-balanced public challenge, Qwen scores 48/90 pairs end to end and 48/64 conditionally; Flash scores 13/90 and 13/61, with `DEMOTE` recalls of 4/30 and 0/30. On the policy graph, local Mistral and Phi-4 score 10/60 and 2/60 pairs; both fail the minimum gate. The deterministic executor defines labels rather than serving as a learned baseline. These results support regression testing of finite typed decision surfaces, not naturalistic policy validity or causal model comparisons.

1 INTRODUCTION

Versioned policies create a specific regression risk at a model-mediated decision boundary. The same artifact may be admitted under one release profile and held under another; one failed verifier field may change the action under a fixed profile. After a policy revision, a route can preserve stale behavior, misread evidence, or fail before returning a valid structured action.

Ordinary accuracy does not separate these cases. A fixed-policy dataset can reward a stable field-to-label shortcut. Evaluating only parseable responses can hide service, syntax, schema, and semantic failures. More fundamentally, a regression suite that covers every edited rule need not cover every behavioral transition induced by the policy revision. We study a bounded question: does a route emit the transition prescribed by a published executable policy for a typed ledger, current state, and requested state? This is conformance to an author-defined executable specification, not a test of scientific truth, software security, policy legitimacy, or general agent competence.

CLAIMPROMOTEBENCH treats the revision, rather than a static dataset, as the source of tests. An *evidence contrast* changes one registered field under a fixed policy. A *policy contrast* holds the complete decision context fixed and changes only the active profile. Every pair changes the prescribed action. This gives an intervention-blind certificate: a route that ignores the intervention cannot pass both members. Pair correctness requires two valid, correct decisions; end-to-end estimates retain every scheduled failure.

The policy contrast turns a revision into a test-generation problem. We execute both policies over a finite authored context grammar, retain their behavioral difference, and compile that set against two adequacy targets: the changed rules witnessed by the suite and the directed action transitions it induces.

Rule-complete tests can omit reachable behavior. The compiler also identifies impossible transitions and selects a cardinality-minimum joint-adequacy suite before any route output is observed.

We instantiate the protocol in two synthetic ontologies. One governs research-evidence states. The other maps SLSA and in-toto attestation fields to an authored software-artifact admission state machine (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). These standards ground the field vocabulary; the authors define the policies and labels. A frozen suite can regression-test a model, prompt, decoder, adapter, or provider against the same contract without assuming a natural input distribution.

Our contributions are:

1. a paired conformance protocol that isolates evidence and active-policy interventions, retains route failure end to end, and certifies the failure of intervention-blind shortcuts;
2. an adaptation of semantic policy differencing to model-route tests, with separate changed-rule and observable action-transition adequacy, impossible-edge certificates, and registered stale-policy mutants;
3. a joint-adequacy minimum-suite formulation and exact dynamic programs that produce a checkable optimum or an edge-count optimality witness; and
4. five frozen evaluations spanning 420 pairs and two authored ontologies, with complete item, pair, action, confusion, and learned-shortcut diagnostics for hosted and pinned local routes.

2 Related Work

Behavioral and conformance testing. CheckList and contrast sets probe local decision boundaries by changing inputs under a fixed target (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing scores relations among transformed executions, often when a direct oracle is unavailable (Chan et al., 1998). Finite-state and ioco testing derive tests from transition specifications, and transition coverage is a standard adequacy criterion (Chow, 1978; Tretmans, 1996); DMN represents executable decision tables (Object Management Group, 2024). ClaimPromoteBench combines these ideas for a narrower setting: an exact oracle exists, the specification itself changes, and the test object is the behavioral difference between two policy versions. A survey of model-mediated software identifies regression and continuous-integration testing as under-supported in part because stable automated oracles are scarce (Gonzalez & El Mezouar, 2026).

Coverage-preserving test-suite reduction also predates our work (Harrold et al., 1993). We use that optimization view only after policy differencing has produced paired model-service tests; the adequacy target jointly covers changed rules and observable action transitions.

Semantic policy differencing is established software-engineering work. Margrave computes witnesses on which two access-control policies disagree; Cirg uses change-impact analysis to generate and reduce policy tests; mutation testing measures access-policy fault detection (Fisler et al., 2005; Martin & Xie, 2007b;a). Semantic differencing has also been applied to evolving combinatorial test models (Tzoref-Brill & Maoz, 2017). Our contribution is not the differencing operation. We turn each witness into two scored model-service executions, retain output-contract failure, and distinguish changed-rule coverage from coverage of the observable four-action changes induced by a revision. The latter includes transitions between non-`ALLOW` actions that a rule-complete suite can miss.

Policy and rule reasoning. Rule-following benchmarks test conversational constraints, unusual instructions, long rule sets, policy comprehension, and request classification (Mu et al., 2024; Zhang et al., 2026; Zhou et al., 2025; Bao et al., 2026; Cisneros-Velarde, 2026). Other systems extract rule flows or internalize policy documents (Yang et al., 2026; Liu et al., 2026a). GuideBench varies domain guidelines, and augmented policy training studies unseen guardrail policies (Diao et al., 2025; Liu et al., 2026b). These works establish policy variation as prior art. They do not compile paired regression tests from the semantic difference of two executable policies or certify coverage of the induced action graph.

SpecEval synthesizes prompts from provider-authored statements and uses model judges to audit adherence; COMPASS generates queries for organization-specific allow and deny lists and judges

free-form responses (Ahmed et al., 2025; Choi et al., 2026). Prompt regression instead asks whether a provider update breaks application behavior (Ma et al., 2024). We hold the route and complete typed context fixed while changing the application policy, use an exact oracle, and score both executions jointly.

Policy-to-Tests translates governance prose into executable rules, while TDAD generates agent tests, optimizes prompts, and evaluates specification evolution with semantic mutants (Datla et al., 2025; Rehan, 2026). CLAIMPROMOTEBENCH starts after that translation: it accepts two deterministic policies and a finite grammar, computes their behavioral difference, and audits changed-rule and induced-transition adequacy. It performs neither policy extraction, prompt optimization, nor judge-based labeling.

Structured output and selective prediction. Grammar-constrained decoding and incremental parsing restrict output form (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench and BFCL evaluate schemas, function selection, and arguments (Geng et al., 2025; Patil et al., 2025). Selective prediction studies risk as a model abstains (Geifman & El-Yaniv, 2017; 2019). We distinguish a valid `HOLD` action from route failure. Malformed or absent output is never treated as abstention and remains incorrect end to end.

Claim–evidence systems. ClaimBench evaluates claim–evidence reasoning in AI papers, while Evidence-Ledger Adjudication, LEDGER, and ClaimLedger organize traceable evidence or claim maintenance (Javaji et al., 2025; Chen et al., 2026; Kim et al., 2026; Elliman, 2026). CLAIMPROMOTEBENCH assumes that typed verifier facts are already registered and tests the subsequent policy-conditioned transition; it neither extracts claim–evidence links nor verifies source truth.

Research agents. ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, computational reproduction, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). CLAIMPROMOTEBENCH tests a smaller component that may appear inside such systems: the transition decision after evidence has already been registered. It does not evaluate evidence discovery or an end-to-end research workflow.

3 TASK AND PROTOCOL

An item is $x = (P, z, s, \tau)$, where P is the active policy, z is a typed evidence ledger, s is the current state, and τ is the requested state. The deterministic executor $\pi_P(x)$ applies first-match precedence

$$\texttt{DEMOTE} > \texttt{ESCALATE} > \texttt{HOLD} > \texttt{ALLOW}.$$

`DEMOTE` lowers a currently elevated state whose recorded basis is invalid. `ESCALATE` preserves state and opens review for an authenticated active conflict. `HOLD` rejects a requested promotion with incomplete prerequisites. `ALLOW` accepts the requested state when no preceding rule matches. The first matching row fixes the action, proposed state, and ordered reason codes.

The evaluated route receives the complete versioned policy contract, one canonical JSON ledger, and passive verifier facts. It must return one JSON object with exactly six fields: `action`, `proposed_state`, `p_admissible`, `used_evidence_ids`, `reason_codes`, and `missing_evidence_ids`. We request no rationale. Exact syntax, schema, and gold-free cross-field checks determine validity. Primary correctness then scores the prescribed action; it does not require exact oracle agreement on probabilities, evidence lists, or reason codes.

Let $v(x)$ indicate a valid response and let $\hat{y}(x)$ denote its action. End-to-end item correctness is

$$C_{\text{item}}(x) = \mathbb{1}[v(x) = 1]\mathbb{1}[\hat{y}(x) = \pi_P(x)].$$

For pair $i = (x_i^a, x_i^b)$, $C_{\text{pair}}(i) = C_{\text{item}}(x_i^a)C_{\text{item}}(x_i^b)$. Decision coverage is the fraction with $v = 1$; pair coverage requires both members to be valid. Conditional accuracies divide correct items or pairs by these coverage denominators. End-to-end accuracies divide by every scheduled item or pair. Thus item accuracy equals decision coverage times conditional item accuracy, with the analogous identity for pairs. Tables encode failure as `N`, not as a fifth policy action. We do not reproduce malformed response contents in the manuscript.

Intervention-blind certificate. **Proposition 1.** Let q denote the controlled coordinate: the active policy P or one registered field of z. For a pair that differs only in q and satisfies $\pi(x_i^a) \neq \pi(x_i^b)$, any deterministic route invariant to q returns the same action for both members. It can answer at most one member correctly and has $C_{\text{pair}}(i) = 0$. Across any such pair set, its end-to-end item accuracy is at most $1/2$ and its pair accuracy is zero. The policy case specializes to a classifier $h(z, s, \tau)$ that cannot observe P; the evidence case covers a route insensitive to the mutated verifier field. *Proof.* The gold actions differ while an invariant route's outputs are equal, so at most one output matches. The certificate depends on the pair construction, not model quality or training procedure.

Policy-difference compilation. For two policies and an authored context grammar $\mathcal{X}$, let $\Delta(P_a, P_b; \mathcal{X})$ contain exactly the contexts $c = (z, s, \tau)$ for which $\pi_{P_a}(c) \neq \pi_{P_b}(c)$. Duplicating a context from Δ and changing only the active profile creates a policy-regression pair. Rule coverage records which policy differences bind in at least one pair. Behavioral coverage instead records directed edges

$$E(P_a, P_b; \mathcal{X}) = \{(\pi_{P_a}(c), \pi_{P_b}(c)) : c \in \Delta(P_a, P_b; \mathcal{X})\}.$$

Neither coverage criterion implies the other. Both are relative to the published context grammar, not to every possible deployment input.

For a test suite T, let $R(T)$ be its witnessed changed rules and $E(T)$ its induced action edges. We report migration adequacy as

$$A(T) = \left(\frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where R_Δ is the authored rule-difference inventory and E_{reach} is obtained by exhaustive execution over the published finite context grammar. Test construction enumerates the grammar, groups contexts by rule witness and ordered action pair, selects the registered balanced contexts, and freezes them before the evaluated route is called. Thus adequacy is established on the oracle side and cannot be chosen from model errors.

Minimum joint-adequacy suites. The compiler is independent of either ontology. Its interface accepts two deterministic policy evaluators, a finite context sequence, stable context identifiers, and authored rule witnesses; it has no SLSA, prompt, model, or provider dependency. Enumerating the behavioral difference costs $O(|\mathcal{X}|(C_a + C_b))$ for policy costs C_a, C_b. **Proposition 2.** Finding a cardinality-minimum joint rule- and edge-complete suite is NP-hard, even when every candidate has the same required edge. The reduction is given in Appendix A. For n candidates, r required rules, and k required edges, our exact subset dynamic program takes $O(n2^{r+k})$ state updates. Each state retains the shortest, then lexicographically first, subset for a coverage mask. An include-or-exclude induction proves optimality at the full mask. A specialized search selects one context per edge while retaining at most 2^r rule masks. If it finds a complete witness, the k-context edge lower bound proves global minimality. Both implementations independently reconstruct the frozen witness from the published 60-pair candidate graph.

3.1 Two authored ontologies

The research-evidence policy uses four state axes: research status, computational support, formal certification, and independent review. Its evidence checks cover dependency closure, runtime completion, scope, freshness, provenance, identity, and contradiction status. The hidden r6 realization and public action-balanced challenge use this same policy but different pair constructions.

The software policy uses release status, accepted build level, and incident-review status. Its fields are grounded in SLSA 1.2 and in-toto Attestation 1.2, including provenance presence, signature verification, subject binding, builder and source bindings, verified build level, and authenticated conflicts (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). The authored state machine maps failed candidate checks to `HOLD`, invalid current bases to `DEMOTE`, and selected authenticated conflicts to `ESCALATE`.

The policy studies publish two counterfactual profiles. `POLICY_BLUE` enforces provenance, subject, builder, and source-ref checks and escalates source-revision or VSA-policy conflicts. `POLICY_AMBER` enforces signature, attester, source-URI, and policy-digest checks and escalates

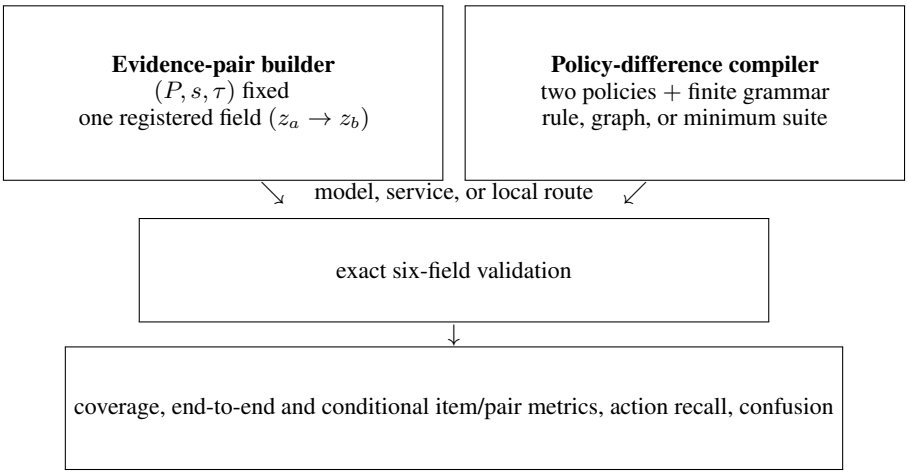


Figure 1: Two controlled paths through one failure-aware evaluator. Policy differencing also emits rule- and edge-coverage certificates, impossible edges, registered mutants, and a minimum smoke suite before route execution.

Table 1: Frozen evaluations. Supports follow ALLOW/HOLD/DEMOTE/ESCALATE. Factors are registered evidence or rule factors except in the last row, where they are reachable directed edges.

Dataset	Contrast	Policy instrument	Pairs	Items	Factors	Support
Research r6 (hidden)	Evidence	Research transition policy	150	300	20	150/134/8/8
Public challenge	Evidence	Research transition policy	90	180	15	90/30/30/30
SLSA portability	Evidence	Artifact-admission policy	60	120	12	60/20/20/20
Policy switch	Policy	BLUE/AMBER	60	120	12	60/20/20/20
Transition graph	Policy	BLUE/AMBER	60	120	10	36/24/24/36

build-parameter or verification-result conflicts. In the rule-stratified suite, each profile has 30 ALLOW decisions and ten for each other action. Both suites give the profiles identical action marginals, so a profile name alone does not predict the action. The profiles and labels are author choices; SLSA and in-toto do not endorse either one.

For example, one pair holds an admitted artifact, its evidence, and its requested state fixed while recording an unrecognized attester. BLUE does not bind that field and prescribes ALLOW; AMBER treats the current admission basis as invalid and prescribes DEMOTE. Only the profile changes.

4 Evaluation Design

Table 1 summarizes the five datasets. The first four suites pair one ALLOW member with a changed action. R6 is a frozen hidden realization with rare DEMOTE and ESCALATE support. The public challenge contains 90 pairs across 15 factors and balances its 90 non-ALLOW members across the other three actions. The transition graph also includes pairs between non-ALLOW actions.

Every model configuration receives one call per item with zero client retries. Hosted studies use two repeated configurations: Qwen3.7-Max with prompt-only JSON and DeepSeek-V4-Flash with a strict JSON schema. Model, provider adapter, and output constraint differ, so between-route comparisons are descriptive. Separately frozen local runs use pinned four-bit Mistral Small 3.2 24B and Phi-4 snapshots with user-only prompts and greedy MLX-LM decoding. They cover the fixed software policy and both policy suites. Quantization, chat templates, and hosted versus local execution preclude causal comparisons.

The research comparator is a multinomial Naive Bayes text classifier trained on 100 public-development items. It is a weak reference. The fixed software policy adds a 500-tree Random Forest trained on 72 development items using flattened model-visible state and verifier fields. Each policy suite trains the same algorithm twice on its development split: the policy-aware version observes the active profile; the profile-blind version does not. Both exclude identifiers, pair meta-

data, evaluation labels, oracle calls, and model outputs. A profile-blind classifier receives identical features for both pair members and therefore cannot answer an action-changing pair completely. Policy-contrast model results use prospectively selected, pinned local routes.

The deterministic executor generates every label and reproduces all items by construction. We call this the *definitional policy ceiling*. It is neither a learned baseline nor external validation of the authored policies.

The evaluation asks whether rule coverage captures the induced action graph, whether a smaller joint-adequacy gate exists, and how model-service routes behave once output-contract failures remain in scope. The first two questions are oracle-side properties; the third is empirical.

5 RESULTS

5.1 MIGRATION ADEQUACY AND MINIMUM GATE

The rule-stratified BLUE→AMBER suite covers all 12 authored profile differences with five contexts each. It contains six admission-centered transitions with ten pairs each: ALLOW↔HOLD, ALLOW↔DEMOTE, and ALLOW↔ESCALATE. The graph suite instead contains six contexts for each reachable directed edge. It covers 10/10 reachable non-self edges and all 12 rules. The remaining directions, HOLD→DEMOTE and DEMOTE→HOLD, are structurally impossible under the published executor: HOLD requires a candidate current state, DEMOTE requires an admitted current state, and pair members share that state. Each profile has the same action support (18/12/12/18 for ALLOW/HOLD/DEMOTE/ESCALATE), blocking a profile-only class prior. The adequacy vectors are $A(T_{\text{rule}}) = (1, .6)$ and $A(T_{\text{graph}}) = (1, 1)$ on the published grammar.

We register one stale-policy mutant m_e for each reachable edge $e = (a, b)$. On an AMBER execution whose correct profile transition is e, m_e returns the stale BLUE action a. A controlled pair kills m_e if and only if it covers e. The rule-stratified suite therefore kills 6/10 registered edge mutants, whereas the graph suite kills 10/10. This oracle-side audit explains the coverage criterion; it is not a model of fault prevalence.

One pair realizes one edge, so ten reachable edges require at least ten pairs. A deterministic search over the six frozen contexts per edge finds a witness that also covers all 12 rules. This is a provably minimum 10-pair, 20-call gate for the joint $(1, 1)$ target on the published candidate pool, an 83.3% reduction from 120 calls. A digest-bound receipt publishes the selected identifiers; no model output enters selection. The search processes edge groups in canonical order and retains the lexicographically first witness for each rule mask.

On the minimum gate, Mistral produced 14/20 valid decisions, 9/20 correct items, and 2/10 correct pairs; on the minimum gate, Phi-4 produced 6/20 valid decisions, 5/20 correct items, and 0/10 correct pairs. Neither route passed the fixed 20/20-item and 10/10-pair rule. This subset reuses terminal records from the full graph run and is not an independent sample.

5.2 COVERAGE AND CORRECTNESS

Table 2 reports every denominator. In r6, Qwen returned 272/300 valid decisions and achieved 258/300 item and 113/150 pair accuracy end to end. The old Flash interface returned HTTP 401 for all 300 requests and produced no model output. A later Flash route used a different service stack; its success establishes only that the later path executed.

The public challenge exposes action errors among valid outputs. Qwen and Flash achieved 132/180 and 90/180 end-to-end item accuracy, with conditional item accuracy 132/148 and 90/146. Pair accuracy fell to 48/90 and 13/90 end to end.

The fixed software policy gives a different decomposition. Flash produced 108/120 valid decisions and 49/60 complete pairs; Qwen produced 88/120 and 28/60. Every valid decision was correct for both hosted routes. Local Mistral produced 94/120 valid and 91/120 correct decisions, yielding 31/60 correct pairs; local Phi-4 produced 102/120 valid and 83/120 correct decisions, yielding 23/60 pairs. The Random Forest reached 120/120 items and 60/60 pairs. The fixed mapping is learnable from the published fields, but route performance still reflects both decision and output-contract errors.

Table 2: Coverage and correctness. E2E retains every scheduled case; conditional columns use valid decisions or complete pairs. N/E denotes an empty conditional denominator.

Configuration	Valid	E2E item	Cond. item	Complete	E2E pair	Cond. pair
Research-evidence policy						
r6 Qwen	272/300	258/300	258/272	123/150	113/150	113/123
r6 text NB	300/300	184/300	184/300	150/150	34/150	34/150
Old r6 Flash interface	0/300	0/300	N/E	0/150	0/150	N/E
Challenge Qwen	148/180	132/180	132/148	64/90	48/90	48/64
Challenge Flash	146/180	90/180	90/146	61/90	13/90	13/61
Fixed software-artifact policy						
SLSA Qwen	88/120	88/120	88/88	28/60	28/60	28/28
SLSA Flash	108/120	108/120	108/108	49/60	49/60	49/49
SLSA Mistral local	94/120	91/120	91/94	34/60	31/60	31/34
SLSA Phi-4 local	102/120	83/120	83/102	42/60	23/60	23/42
SLSA structured RF	120/120	120/120	120/120	60/60	60/60	60/60
Counterfactual active-policy switch						
Switch Mistral local	77/120	38/120	38/77	33/60	6/60	6/33
Switch Phi-4 local	59/120	46/120	46/59	14/60	4/60	4/14
Switch policy-aware RF	120/120	120/120	120/120	60/60	60/60	60/60
Switch profile-blind RF	120/120	60/120	60/120	60/60	0/60	0/60
Reachable policy-transition graph						
Graph Mistral local	71/120	40/120	40/71	31/60	10/60	10/31
Graph Phi-4 local	43/120	27/120	27/43	8/60	2/60	2/8
Graph policy-aware RF	120/120	107/120	107/120	60/60	47/60	47/60
Graph profile-blind RF	120/120	57/120	57/120	60/60	0/60	0/60

On the rule-stratified policy contrast, local Mistral and Phi-4 produced 77/120 and 59/120 valid decisions, achieved 38/120 and 46/120 item accuracy, and completed 6/60 and 4/60 pairs correctly. Its policy-aware Random Forest achieved 120/120 items and 60/60 pairs, while its profile-blind counterpart achieved 60/120 items and 0/60 pairs. On the transition graph, Mistral produced 71/120 valid and 40/120 correct decisions, with 10/60 correct pairs; Phi-4 produced 43/120, 27/120, and 2/60. Conditional pair accuracy was 10/31 and 2/8. The policy-aware and profile-blind Random Forests reached 47/60 and 0/60 pairs. Profile-blind models are registered shortcut controls, not competitive baselines.

5.3 Action-level diagnostics

R6 contains only eight `DEMOTE` and eight `ESCALATE` items, and the weak text baseline recovers neither class. The public challenge raises each non-`ALLOW` support to 30: Qwen recovers 4/30 `DEMOTE` cases and Flash recovers 0/30. On the SLSA contrast, Qwen recovers 1/20 `DEMOTE` and 7/20 `ESCALATE` cases; Flash recovers 15/20 and 17/20. Both local configurations recover 0/20 `DEMOTE` cases on the fixed policy and policy switch. On the graph, both again recover 0/24. Mistral recovers 9/36 `ALLOW`, 18/24 `HOLD`, and 13/36 `ESCALATE` items; Phi-4 recovers 21/36, 5/24, and 1/36. Appendix B reports every action recall and confusion cell, including no-valid-decision counts.

5.4 Factor diagnostics

Pair outcomes localize different failure modes. R6 Qwen scores 0/7 pairs on dependency-statement changes and 0/8 on current-state binding; seven of 20 factors are perfect. Challenge Qwen has three zero-accuracy factors and two perfect factors, while challenge Flash has 11 zero-accuracy factors. Each factor has only six to eight pairs, so these are diagnostics rather than stable rankings.

On the SLSA contrast, conditional pair accuracy is 1.0 for every covered factor in both hosted routes. The local routes add action errors after validation: Mistral reaches 31/34 conditional pair accuracy, while Phi-4 reaches 23/42. On the policy switch, Mistral obtains five of its six correct pairs from one authenticated-conflict factor; Phi-4's four correct pairs occur in three candidate-admission factors.

On the graph, Mistral's ten correct pairs occupy three edges, including all six `HOLD`→`ESCALATE` pairs; Phi-4's two correct pairs both lie on `ALLOW`→`HOLD`. Mistral has no complete pair on two edges and Phi-4 on seven. These cells diagnose the frozen routes rather than estimate edge prevalence. The artifact retains every factor numerator, denominator, failure category, and item-level scoring record.

6 DISCUSSION

Evidence and policy contrasts test different shortcuts. The fixed SLSA Random Forest is perfect on held-out templates, but this does not establish policy conditioning. On both policy suites, removing the profile makes pair members feature-identical but label-different and yields 0/60 pair accuracy. Policy contrasts instead test whether the action changes with the published policy while evidence remains fixed.

Policy revisions yield regression tests. Rule coverage misses four reachable non-`ALLOW` edges; the graph covers all ten. The registered mutants mirror this distinction by construction, so they validate adequacy rather than estimate fault prevalence. The 20-call minimum gate does not replace the 60-pair diagnostic, which retains repeated contexts and factor localization. It rejects both local routes, but two reused subsets do not estimate the gate's sensitivity or specificity. A policy edit regenerates the difference set and receipt, whereas a route edit reruns frozen pairs. This separates missing coverage, invalid output, and wrong actions. The compiler applies to finite typed decision surfaces, but we validate it only on two authored ontologies; admission, escalation, and authorization remain prospective uses.

Failure-aware scoring changes the diagnosis. The SLSA hosted routes make no action mistakes after validation, yet their end-to-end pair accuracies are 49/60 and 28/60. Conditional accuracy alone would report 1.0 and erase 43 incomplete pairs across the two routes. The local routes show the complementary case on the same frozen corpus: some valid outputs choose the wrong action. Reporting coverage, conditional correctness, and end-to-end correctness distinguishes these layers. On the balanced challenge, Qwen's 48/64 conditional pair score becomes 48/90 end to end; Flash's 13/61 becomes 13/90. Their 4/30 and 0/30 `DEMOTE` recalls expose a class asymmetry hidden by r6.

The protocol evaluates a model boundary; it does not advocate one. When policy and evidence are fully structured, a deterministic executor should enforce the transition. The learned routes emulate systems that already place a model at this tool-facing boundary. CLAIMPROMOTEBENCH tests that boundary; it does not show that a model is preferable to the executor.

Route differences are descriptive. The old r6 Flash interface produced 300 HTTP 401 responses; later Flash configurations used a different adapter and output contract. This proves neither model improvement nor an ablation effect. Hosted and local routes likewise differ in weights, quantization, prompts, adapters, and constraints.

7 LIMITATIONS

All labels come from authored synthetic policies. Compilation assumes two deterministic policies, authored rule witnesses, and a finite grammar; it neither extracts rules from prose nor exhausts deployment behavior. Action edges are recomputed, but incorrect witnesses would invalidate the rule-coverage certificate. SLSA and in-toto ground the software fields but do not endorse our state machine. Two ontologies do not define a policy population, and the profiles are fixed instruments rather than unseen-policy synthesis.

R6 and the public challenge share one ontology and mostly the same templates; the challenge is public robustness evidence, not hidden-test generalization. Canonical ledgers leave extraction, ambiguity, policy retrieval, and downstream outcomes untested. Reviewing 12 source-traceable CORE-Bench candidates created zero cases and zero model calls because time-stamped artifacts and provenance were missing. Naturalistic validity therefore remains untested. No human annotators or agreement statistic are claimed.

Each configuration receives one call per item with zero retries. The results do not estimate sampling variance, temporal stability, retry policies, or a model population. Primary conformance is action-level after structural validation, not six-field exact match. Local routes use four-bit snapshots on one Apple Silicon stack; route differences are not causal.

The research ontology has only a weak Naive Bayes comparator. The Random Forest's fixed-policy result exposes direct label encoding, and its policy-aware graph result is 107/120. Profile ablations

close one shortcut but do not establish rule following under paraphrase, long policies, adversarial evidence, or unseen profile composition.

The edge-local mutants cover only their registered family, and minimum-suite optimality is candidate-pool-relative. The minimum-gate scores reuse the graph records and are not an independent replication. Quota-confounded hosted policy-switch studies and a 40-cell hosted smoke are excluded in full; no successful hosted row, conditional subset, or hosted model rows enter the paper. Invalid response contents are omitted from the paper and anonymous release, but their counts remain in end-to-end denominators and auditable records.

8 Reproducibility Statement

The artifact binds policies, source standards, generators, datasets, prompt packets, model revisions, schedules, one-call terminal records, analyses, and privacy receipts. Validators recompute every reported metric and confusion cell from item records. The anonymous derivative removes raw response text, free-form errors, provider identifiers, hidden reasoning, account identity, and private commit identity; its manifest derives file counts from the generated tree. Excluded protocol attempts are represented only by redacted receipts and contribute no model rows. Completed one-shot studies are never rerun.

9 AI Assistance Disclosure

Generative AI tools assisted literature search, planning, code and benchmark drafting, debugging, analysis checks, and manuscript editing. The author chose the research question, policies, protocols, exclusions, analyses, and claims; verified the datasets, results, artifacts, and references; and is responsible for the manuscript. AI systems are neither authors nor human annotators.

10 Conclusion

ClaimPromoteBench turns executable policy revisions into paired route tests with two oracle-side adequacy axes. On the authored migration, rule coverage misses four of ten reachable action edges; the exact minimum joint gate retains all 12 changed rules in 20 calls. Local graph routes reach 10/60 and 2/60 pairs and both fail that gate. Across r6, the public challenge, and the software-policy studies, gaps between end-to-end and conditional scores and low `DEMOTE` recall separate route validity from action conformance. These results support regression testing of finite typed policy surfaces, not naturalistic policy validity or causal model comparisons.

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A Compiler Guarantees

Proof of Proposition 2. Take a Set Cover instance with universe U and subsets $S_1, \ldots, S_n$. Create one required rule for each element of U and one candidate context for each S_i whose rule-witness set is S_i. Give every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover U. A minimum joint-adequacy suite therefore gives a minimum set cover. The construction is polynomial.

Exact dynamic program. Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first j candidates, the table stores, for every reachable mask, the shortest subset that realizes that mask, breaking ties by context identifier. This invariant holds initially for the empty mask. For candidate $j + 1$, the exclude transition preserves every prior state and the include transition unions its coverage mask with each prior mask. Keeping the shorter, then lexicographically first, representative preserves the invariant. At $j = n$, the representative for the full mask is therefore a cardinality-minimum joint-complete suite. The algorithm performs $O(n2^{r+k})$ state updates and stores at most 2^{r+k} states.

Edge lower-bound certificate. Each policy-contrast pair realizes one directed action edge. Covering k distinct required edges therefore needs at least k pairs. The specialized search chooses one candidate from each edge group while tracking rule masks. If its full-rule state exists, the returned k-pair suite meets the lower bound and is globally minimum. The generic and specialized implementations both reconstruct the frozen ten-pair witness. The exact implementation also matches exhaustive subset enumeration on 64 deterministic small candidate pools.

B Complete Action-Level Results

Table 3: End-to-end recall by gold action. Every no-valid-decision cell is a miss. Bal. is the unweighted mean over the four recalls.

Configuration	ALLOW	HOLD	DEMOTE	ESCALATE	Bal.
r6 Qwen	.9533	.8209	.0000	.6250	.5998
r6 text NB	.5000	.8134	.0000	.0000	.3284
Old r6 Flash interface	.0000	.0000	.0000	.0000	.0000
Challenge Qwen	.8778	.8667	.1333	.7667	.6611
Challenge Flash	.7556	.3000	.0000	.4333	.3722
SLSA Qwen	1.0000	1.0000	.0500	.3500	.6000
SLSA Flash	.9833	.8500	.7500	.8500	.8583
SLSA Mistral local	1.0000	.9500	.0000	.6000	.6375
SLSA Phi-4 local	1.0000	.2000	.0000	.9500	.5375
SLSA structured RF	1.0000	1.0000	1.0000	1.0000	1.0000
Switch Mistral local	.2667	.6000	.0000	.5000	.3417
Switch Phi-4 local	.6333	.3000	.0000	.1000	.2583
Switch policy-aware RF	1.0000	1.0000	1.0000	1.0000	1.0000
Switch profile-blind RF	.0000	1.0000	1.0000	1.0000	.7500
Graph Mistral local	.2500	.7500	.0000	.3611	.3403
Graph Phi-4 local	.5833	.2083	.0000	.0278	.2049
Graph policy-aware RF	.8889	.7917	1.0000	.8889	.8924
Graph profile-blind RF	.2500	.5833	1.0000	.2778	.5278

Table 4: Conditional recall by gold action among valid decisions. N/E denotes no valid decision for that gold action. This table must be read with the end-to-end recalls above.

Configuration	ALLOW	HOLD	DEMOTE	ESCALATE
r6 Qwen	1.0000	.9483	.0000	1.0000
r6 text NB	.5000	.8134	.0000	.0000
Old r6 Flash interface	N/E	N/E	N/E	N/E
Challenge Qwen	1.0000	1.0000	.2000	1.0000
Challenge Flash	.8500	.4091	.0000	.8125
SLSA Qwen	1.0000	1.0000	1.0000	1.0000
SLSA Flash	1.0000	1.0000	1.0000	1.0000
SLSA Mistral local	1.0000	.9500	.0000	1.0000
SLSA Phi-4 local	1.0000	.2222	.0000	.9500
SLSA structured RF	1.0000	1.0000	1.0000	1.0000
Switch Mistral local	.3636	.6000	N/E	.7692
Switch Phi-4 local	1.0000	1.0000	.0000	1.0000
Switch policy-aware RF	1.0000	1.0000	1.0000	1.0000
Switch profile-blind RF	.0000	1.0000	1.0000	1.0000
Graph Mistral local	.3462	.7500	.0000	.7222
Graph Phi-4 local	1.0000	.8333	.0000	1.0000
Graph policy-aware RF	.8889	.7917	1.0000	.8889
Graph profile-blind RF	.2500	.5833	1.0000	.2778

Each confusion entry follows predicted-column order [ALLOW, HOLD, DEMOTE, ESCALATE, N], where N means no valid decision.

Table 5: Full end-to-end confusion matrices in compact row form. Columns after configuration correspond to gold ALLOW, HOLD, DEMOTE, and ESCALATE.

Configuration	Gold ALLOW	Gold HOLD	Gold DEMOTE	Gold ESCALATE
r6 Qwen	143/0/0/0/7	4/110/0/2/18	8/0/0/0/0	0/0/0/5/3
r6 text NB	75/75/0/0/0	25/109/0/0/0	8/0/0/0/0	8/0/0/0/0
Old r6 Flash interface	0/0/0/0/150	0/0/0/0/134	0/0/0/0/8	0/0/0/0/8
Challenge Qwen	79/0/0/0/11	0/26/0/0/4	8/4/4/4/10	0/0/0/23/7
Challenge Flash	68/4/0/8/10	13/9/0/0/8	25/3/0/0/2	0/3/0/13/14
SLSA Qwen	60/0/0/0/0	0/20/0/0/0	0/0/1/0/19	0/0/0/7/13
SLSA Flash	59/0/0/0/1	0/17/0/0/3	0/0/15/0/5	0/0/0/17/3
SLSA Mistral local	60/0/0/0/0	1/19/0/0/0	0/2/0/0/18	0/0/0/12/8
SLSA Phi-4 local	60/0/0/0/0	14/4/0/0/2	4/0/0/0/16	1/0/0/19/0
SLSA structured RF	60/0/0/0/0	0/20/0/0/0	0/0/20/0/0	0/0/0/20/0
Switch Mistral local	16/13/0/15/16	8/12/0/0/0	0/0/0/0/20	3/0/0/10/7
Switch Phi-4 local	38/0/0/0/22	0/6/0/0/14	4/9/0/0/7	0/0/0/2/18
Switch policy-aware RF	60/0/0/0/0	0/20/0/0/0	0/0/20/0/0	0/0/0/20/0
Switch profile-blind RF	0/20/20/20/0	0/20/0/0/0	0/0/20/0/0	0/0/0/20/0
Graph Mistral local	9/8/0/9/10	2/18/0/4/0	3/0/0/0/21	2/3/0/13/18
Graph Phi-4 local	21/0/0/0/15	1/5/0/0/18	3/12/0/0/9	0/0/0/1/35
Graph policy-aware RF	32/0/3/1/0	5/19/0/0/0	0/0/24/0/0	3/0/1/32/0
Graph profile-blind RF	9/8/12/7/0	7/14/0/3/0	0/0/24/0/0	8/6/12/10/0